SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

BizPilot — AI Business Assistant for Small Businesses

Project Type: AI-Native Multi-Agent Business Assistant

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements of BizPilot, an AI-powered business assistant designed for small-business owners in India.

The document specifies the system's objectives, scope, user roles, core features, interfaces, data requirements, security constraints, AI-related requirements, and acceptance criteria.

It will serve as a common reference for Team Nexus throughout the software development lifecycle.

1.2 Project Overview

BizPilot is a conversational AI business assistant that helps small-business owners manage routine business operations through a single Hindi/English interface.

The system uses a Manager/Orchestrator Agent to route requests to specialized agents responsible for customer queries, inventory management, billing, and business insights.

Rather than functioning as a general-purpose chatbot, BizPilot uses verified business data, controlled tools, configurable business rules, and human approval for sensitive operations.

1.3 Problem Statement

Small-business owners often manage customer communication, inventory, billing, and business decisions manually. This creates several operational challenges:

- Delayed or missed customer responses result in lost sales.
- Inaccurate inventory records can lead to unexpected stockouts or excess unsold inventory.
- Manual invoice preparation introduces calculation or tax-related errors.
- Limited business analytics make it difficult to identify sales trends and product performance.
- English-first software and complex interfaces make it difficult for users with limited technical experience.
- Hiring additional staff to manage repetitive tasks is financially impractical for small businesses.

BizPilot aims to address these problems through an affordable, conversational, AI-assisted system with appropriate safeguards.

1.4 Project Objectives

The primary objectives of BizPilot are:

- Provide a simple Hindi/English interface for business operations.
- Automate routine customer queries using verified prices, stock levels, and order information.
- Monitor inventory and identify low-stock and slow-moving products.
- Generate invoices using verified product and tax data.
- Produce daily and weekly business summaries.
- Coordinate specialized AI agents through a controlled orchestration workflow.
- Protect business information through access controls and secure logging.
- Evaluate agent performance, response quality, safety, and operating cost.

1.5 Scope

In scope

- Owner authentication and business profile management.
- Inventory management.
- Customer query handling.
- Hindi and English text interaction, including Hinglish where supported.
- Agent routing and inter-agent communication.
- Low-stock alerts and reorder suggestions.
- Invoice generation and verified tax-rule application.
- Daily and weekly sales summaries.
- Configurable approval rules.
- AI evaluation, logging, monitoring, and basic cost tracking.
- A web-based owner dashboard.

- Messaging integration were supported by the available APIs and project environment.

Future Scope

- Fully autonomous supplier payments or purchases without owner approval.
- Replacing a qualified chartered accountant for complex tax decisions.
- Guaranteeing legal or tax compliance without verified rules and appropriate review.
- Providing lending, investment, or other unrelated financial advice.
- Supporting every business type, language, tax scenario, or external messaging platform.
- Making business decisions using fabricated or unverified data.

1.6 Stakeholders

This document is intended for:

- Project mentors and EPAM evaluators.
- Business and product analysts.
- Frontend and backend developer.
- AI/ML and agent-workflow developer.
- Database and integration developer.
- Quality assurance and security tester.

1.7 Definitions and Abbreviations

- **SRS:** Software Requirements Specification.
- **AI:** Artificial Intelligence.
- **LLM:** Large Language Model.
- **Agent:** A software component assigned a specific task and permitted to use defined tools.
- **Orchestrator:** Component responsible for classifying requests and coordinating agent execution.
- **API:** Application Programming Interface.
- **MCP:** Model Context Protocol, used to expose tools and resources to compatible AI clients.

- **GST:** Goods and Services Tax.
- **GSTIN:** Goods and Services Tax Identification Number.
- **PII:** Personally Identifiable Information.
- **RAG:** Retrieval-Augmented Generation.
- **Evaluation (Eval):** A test used to measure the correctness, reliability, or safety of an AI system.
- **Guardrail:** A rule or technical control that restricts unsafe or unauthorized actions.
- **Human-in-the-loop:** A workflow in which a person reviews or approves specified actions before execution.

1.8 References

- BizPilot Project Idea presentation.
- BizPilot project problem statement.
- Applicable official GST guidance and verified tax-rule sources, to be identified during implementation.

2. Overall Description

2.1 Product Perspective

BizPilot is a multi-agent business application consisting of a user interface, backend services, a shared business database, an AI orchestration layer, specialized agents, and integrations with external services.

The proposed architecture is:

->User Interface / Messaging Channel
→ Backend API and Authentication
→ Manager/Orchestrator Agent
→ Specialized Business Agents
→ Authorized Tools and Verified Business Data
→ Result Validation and Approval
→ Response or Business Action
→ Audit Log and Dashboard Update

The system must treat the database and deterministic business tools as the source of truth for prices, inventory quantities, order records, and invoice calculations. The language model may

interpret requests and prepare explanations, but it must not independently invent authoritative business values.

2.2 User Classes

A. Business Owner / Administrator

The **owner** is the **primary user** and is responsible for managing the business.

The owner shall be able to:

- Configure business details and operating rules.
- Add, update, and view products and stock.
- Review orders and invoices.
- Configure low-stock and approval thresholds.
- Review sales summaries and stock alerts.
- Access authorized logs and evaluation information.

B. Customer

The customer interacts with BizPilot to obtain business-related information.

The customer shall be able to:

- Ask about product availability and selling prices.
- Ask permitted questions about orders.
- Receive responses in a supported language.

C. Project Administrator / Authorized Operator

An authorized operator may manage system configuration, monitor errors, and review operational logs, subject to role-based permissions.

Access to business data must be restricted to the operator's authorized responsibilities.

2.3 Operating Environment

The proposed initial environment includes:

- A modern desktop or mobile web browser.
- A responsive owner dashboard.

- A backend API service.
- A relational database.
- An LLM provider and agent orchestration framework.
- An internet connection for hosted AI and external integrations.

2.4 Design and Implementation Constraints

- The system must operate within the budget and infrastructure available to the project team.
- External AI services may impose rate limits, usage costs, or availability constraints.
- Messaging APIs may require registration, permissions, and approval.
- AI-generated content must not bypass deterministic business rules.
- GST calculations must use verified, applicable tax rules and validated inputs.
- Each agent must receive only the information and tools required for its assigned task.
- Sensitive actions must follow configurable approval rules.
- The initial implementation may limit the number of supported languages, business categories, and integrations.

2.5 User Expectations

The system should be easy to learn, affordable to operate, transparent about its actions, and understandable to users with limited technical experience.

It should clearly distinguish confirmed facts, recommendations, pending approvals, and failed operations.

3. Functional requirements

3.1 Manager / Orchestrator Agent

01 — Request Reception
The system shall accept a business-related request from an enabled user interface or messaging channel.

02 — Intent Classification
The Manager Agent shall classify the request into a supported category, such as customer enquiry, inventory management, billing, sales insights, or general business assistance.

03 — Agent Routing
The Manager Agent shall route each request to the appropriate specialized agent based on the identified intent and the user's permissions.

04 — Multi-Agent Coordination
The system shall allow an agent to request information or an authorized operation from another agent through a defined structured message format.

05 — Context Management
The system shall provide each agent only the relevant business information required to complete its task.

06 — Result Validation
The system shall validate tool results and applicable business rules before presenting a confirmed result or executing an action.

07 — Unsupported Requests
If a request is unsupported or cannot be processed reliably, the system shall ask for clarification or explain the limitation instead of inventing an answer.

08 — Execution Tracking
The system shall record the request identifier, selected agent, relevant tool calls, execution outcome, and error status, while masking sensitive information where required.

Example:

Input: "Parle-G available hai kya?"

- The Manager Agent identifies a product-availability query.
- The request is routed to the Customer Agent.
- The Customer Agent requests stock and price information from the Inventory Agent or an authorized product lookup tool.
- The system verifies the returned values.
- The Customer Agent responds in the supported language.

3.2 Customer Agent

01 — Product Availability
The Customer Agent shall retrieve the current recorded stock status before confirming product availability.

02 — Price Information
The Customer Agent shall return the selling price recorded in the business database and shall not invent or independently alter prices.

03 — Frequently Asked Questions
The Customer Agent shall answer supported questions using approved business information, such as shop timings and return policies, when such information has been configured.

04 — Language Support
The Customer Agent shall generate responses in the supported language selected or used by the customer.

05 — Unavailable Information
If the requested information is missing, outdated, or unavailable, the agent shall communicate that limitation rather than make an unsupported claim.

06 — Restricted Information
The Customer Agent shall not disclose internal cost prices, private owner data, credentials.

3.3 Inventory Agent

01 — Product Records
The system shall allow authorized users to add, view, and update product records, including product name, identifier, selling price, available quantity, and configured reorder threshold.

02 — Stock Updates
The system shall update inventory records after a confirmed sale, approved return, or authorized manual stock adjustment.

03 — Low-Stock Alerts
The system shall identify products whose recorded quantity is at or below the configured reorder threshold and notify the owner through an enabled channel.

04 — Availability Verification
The Inventory Agent shall return the recorded available quantity and stock status for a requested product.

05 — Reorder Suggestions
The system shall generate reorder suggestions based on available inventory and configured business rules.

06 — Slow-Moving Inventory
Where sufficient sales history exists, the system shall identify products with low or declining sales according to a configurable analysis period.

07 — Invalid Quantities
The system shall reject invalid stock adjustments, such as negative quantities where not permitted, and shall explain the validation failure.

08 — Inventory Consistency
The system shall prevent an order from being confirmed when the available stock is insufficient, unless a separately defined and authorized backorder process is supported.

09 — Audit Trail
The system shall record inventory changes, their timestamps, and their sources, subject to the applicable access-control policy.

3.4 Billing Agent

01 — Invoice Creation
The Billing Agent shall generate an invoice using verified order details, product prices, quantities, and configured business information.

02 — Calculation Validation
The system shall calculate line totals, applicable taxes, discounts, and the final amount using deterministic calculation logic rather than relying solely on LLM-generated arithmetic.

03 — Tax Rule Verification
The system shall apply a verified and applicable tax rule only when the necessary information is available. It shall not guess a GST rate.

04 — Required Information
The system shall validate the invoice fields required for the applicable transaction before finalizing the invoice.

05 — Invoice Output
The system shall provide a readable invoice, with PDF output where supported by the implementation.

06 — Invoice Consistency
The system shall ensure that the invoice total matches the validated line items, discounts, and tax calculations.

07 — Approval Controls
The system shall obtain owner approval before executing any billing-related action that falls under a configured approval rule.

08 — Missing or Ambiguous Tax Information
If the available information is insufficient to determine the applicable tax treatment, the system shall request the missing information or flag the invoice for review.

09 — Invoice Records
The system shall store invoice identifiers, associated orders, calculation results, timestamps, and status.

10 — Invoice Corrections
Corrections, cancellations, and refunds shall follow explicitly defined authorization and record-keeping rules.

3.5 Insights Agent

01 — Daily Sales Summary
The Insights Agent shall generate a daily summary of recorded sales for a specified business date.

02 — Weekly Sales Summary
The system shall generate a weekly sales summary when sufficient transaction records are available.

03 — Best-Selling Products
The system shall identify top-selling products for a selected reporting period based on recorded transaction quantities or sales values.

04 — Sales Trends
The system shall compare sales across selected periods when sufficient historical data is available.

05 — Stock Risk Insights
The system shall highlight products that are low in stock or at risk of stockout based on available inventory and sales information.

06 — Profit Estimates
Where verified cost-price and transaction data are available, the system may calculate a clearly defined gross-profit estimate.

07 — Missing Data Disclosure
The agent shall disclose when incomplete transaction or cost data prevents reliable reporting.

08 — Summary Delivery
The system shall display the summary on the owner dashboard and support delivery through an enabled notification channel where implemented.

09 — Traceable Results
Reported figures shall be derived from stored transaction records and reproducible calculation logic.

3.6 Owner Approval and Guardrails

01 — Tool Authorization
Each agent shall be restricted to its approved tools and permitted operations.

02 — Failure-Safe Behavior
If approval status, stock information, tax rules, or another critical input cannot be verified, the system shall not silently execute the affected high-risk action.

03 — Explainable Decisions
Where practical, the system shall tell the owner why an action was blocked, flagged, or sent for approval.

4. Non-functional requirements (NFRs)

4.1 Security and privacy requirements

- Authentication credentials and API keys shall be stored securely and shall not be committed to source control.
- Authorization checks shall be enforced by the backend, not just by hiding frontend buttons.
- Business records belonging to one shop shall not be accessible to another shop without explicit authorization.
- Logs shall use redaction or masking for sensitive fields.
- The system shall validate and restrict tool inputs before executing database changes, sending messages, or generating invoices.
- Customer messages shall be treated as untrusted input, even when they contain instructions directed at the AI.

- Agents shall receive only the data and permissions required for their assigned tasks.
- The system shall maintain an audit trail for sensitive actions and owner approvals.
- Errors shall not expose credentials, internal prompts, or private business records.

4.2 AI-specific requirements

Because BizPilot is a multi-agent AI project:

- Each agent shall have a documented purpose, permitted tools, and operating instructions.
- Agent-to-agent communication shall use defined input and output schemas.
- Structured outputs shall be validated before being passed to business tools.
- Prompt and instruction evaluations shall cover normal, ambiguous, unsupported, and adversarial queries.
- The evaluation dataset should include Hindi, English, and Hinglish examples.
- Failed operations shall be logged, and retry policies shall be bounded to prevent repeated uncontrolled execution.
- Repeated requests should not create duplicate invoices when the same operation is retried.

5. External interfaces and system architecture

5.1 User interface requirements

The owner dashboard should provide:

- Login and business profile management.
- A conversational interface.
- Product and inventory management.
- Low-stock notifications.
- Order and invoice history.
- Pending approval requests.
- Daily and weekly reports.
- An error or activity view for authorized users.

5.2 Software interfaces

Interface	Purpose
Frontend to backend API	Submit requests and display authorized results.
Backend to database	Read and update products, inventory, orders, invoices, and business rules.
Backend to LLM	Interpret natural-language requests and generate structured responses.
Orchestrator to agents	Route tasks and exchange structured results.
Agents to business tools	Perform verified lookups, calculations, and permitted actions.
Backend to messaging provider	Send and receive messages when integration is enabled.
Monitoring and evaluation tools	Capture execution traces, errors, latency, and evaluation results.

5.3 Data requirements

Entity	Example field
Business	Business ID, name, contact details, settings, configured rules
Product	Product ID, name, selling price, cost price where applicable, tax classification
Inventory	Product ID, available quantity, reorder threshold, last updated timestamp
Order	Order ID, items, quantity, total, status, timestamps
Invoice	Invoice ID, order reference, line totals, tax breakdown, final amount, status
Approval	Approval ID, action reference, requested amount, decision, approver, timestamp
Agent execution	Request ID, agent name, tool outcome, latency, error status
Sales record	Transaction ID, date, product lines, quantities, amounts

6. Use Cases and Acceptance Criteria — Summary

This section defines how users interact with the BizPilot system and the conditions each feature must satisfy.

1. Product Availability — The Customer Agent provides accurate product availability and pricing using recorded inventory data, without inventing missing information.

2. Inventory Monitoring — The Inventory Agent detects low-stock products based on configured thresholds and alerts the business owner. Reorders require approval where applicable.

3. Invoice Generation — The Billing Agent generates invoices using validated order data and deterministic calculations. Incomplete or invalid invoices are held for review.

4. Daily Sales Summary — The Insights Agent generates sales reports from recorded transactions, ensuring accurate totals and clearly identifying profit estimates when cost data is available.

Overall objective: Ensure that BizPilot provides accurate, traceable, and reliable business operations with proper validation, error handling, and approval controls.

7. Testing and project acceptance

7.1 Acceptance criteria

Test area	**Proposed success condition**
Request routing	Each supported test query is routed to the intended agent.
Customer responses	Prices and availability match verified test records.
Inventory	Low-stock alerts correctly reflect configured thresholds.
Billing	Invoice calculations match independently calculated expected results across the agreed test cases.
Access control	Unauthorized attempts to access restricted records are denied.
Language handling	Agreed Hindi and English test cases produce understandable, relevant responses.
Error handling	Failed tool calls do not produce fabricated success messages or unintended duplicate actions.
Insights	Sales summaries match the underlying test transactions.
AI evaluation	The team records accuracy, safety, routing and failure results on an agreed evaluation dataset.

7.2 Success metrics

BizPilot's success will be evaluated through a pilot by measuring response speed, accuracy, automation efficiency, and security.

- Performance: Median and 95th-percentile response time.
- AI accuracy: Correct agent routing and accurate stock and price responses.
- Billing reliability: Invoice calculation error rate.
- Security: Unauthorized actions blocked and approval-required actions correctly held.
- Operational reliability: Number of duplicate operations.
- Business impact: Owner time saved on repetitive tasks.
- Cost efficiency: AI cost per conversation, where measurable.

8. Risks, limitations and future enhancements

Risks

Risk	Mitigation
Incorrect inventory data	Timestamp updates, validate changes, and provide manual correction facilities.
Incorrect tax information	Use verified tax rules and require review when information is insufficient.
AI hallucinations	Retrieve verified data and validate outputs before executing actions.
Prompt injection	Enforce backend permissions, tool restrictions, and business rules independently of model instructions.
External API failures	Use bounded retries, error reporting, and safe recovery mechanisms.
Duplicate invoices	Use unique operation identifiers and idempotent processing where appropriate.
High AI operating costs	Track usage, cache suitable responses, and select models according to task complexity.
Incomplete adoption by shopkeeper	Provide a simple interface, clear explanations, and owner control over actions

8.1 Future enhancements

- Voice-based Hindi interaction.
- Additional Indian languages.
- Supplier integrations and purchase-order workflows.

- More advanced demand forecasting.
- Automated recurring business reports.
- Additional messaging platforms.
- Multi-store inventory management.
- Improved analytics and configurable dashboards.